

INTERNSHIP REPORT

On

Diwali Sales Analysis Using Python

Submitted To

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B.Tech CSE



By

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CANDIDATE'S DECLARATION

I, Harsh Raj, Roll No. CS-2341780, do hereby declare that the internship report titled “Diwali Sales Analysis Using Python” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references used have been duly quoted and acknowledged, and I have not taken as such any material from any other source without proper citation. I affirm that this report is my own original work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout this internship. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

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I also express my gratitude to my parents for their blessings and constant encouragement.

Signature: _____

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INTERNSHIP COMPLETION CERTIFICATE



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4. PROJECT DESCRIPTION

4.1 Introduction

Festive sales, especially Diwali sales in India, significantly influence annual revenue for e-commerce and retail businesses. Understanding customer trends during this period is crucial for planning marketing, inventory, and pricing strategies. This project performs an in-depth analysis of a Diwali sales dataset using Python-based data analytics techniques to extract meaningful patterns and identify the key customer segments that contribute most to overall sales.

The study begins with thorough data cleaning and preprocessing to remove inconsistencies, null values, and irrelevant fields, ensuring the dataset is suitable for analysis. Exploratory Data Analysis (EDA) is then conducted using data visualization libraries such as Matplotlib and Seaborn to examine trends related to gender, age group, occupation, state-wise contribution, and product preferences.

4.2 Organization Profile

This internship was undertaken under the “Data Science and Machine Learning” track, conducted in collaboration with the All India Council for Technical Education (AICTE), UniConverge Technologies, and UpSkill Campus. The program, based in Noida, provided hands-on training along with real industry project work, combining conceptual learning with practical, dataset-driven assignments. The internship was carried out from 24-07-2026 to 24-08-2026, during which the Diwali Sales Analysis project was completed under the guidance of the program mentors, alongside academic supervision by Mr. Ashis Ratha, School of Computer Science and Engineering, IILM University.

4.3 Problem Statement

Businesses often lack structured insights about customer buying patterns during festive periods. Without data-driven decisions, they miss opportunities for targeted marketing and sales optimization. This project addresses the need for a systematic, data-driven analysis of Diwali sales data to uncover actionable customer insights.

4.4 Project Objectives

- To clean and preprocess raw e-commerce sales data.

- To perform Exploratory Data Analysis (EDA) using Python.
- To identify top-performing demographics and product categories.
- To derive insights useful for marketing and business growth.

4.5 Scope of the Project

The scope of this project is limited to the analysis of a single Diwali sales dataset (CSV format) using Python. It covers data cleaning, exploratory analysis across gender, age group, state, occupation, and product category dimensions, and the derivation of business insights. The project does not include live data integration, predictive modelling, or deployment, though these are identified as areas for future enhancement.

4.6 Technologies and Tools Used

Table 4.1: Tools and Libraries used in the project

Tool	Purpose
Python	Primary programming language
Pandas	Data cleaning & manipulation
NumPy	Numerical computations
Matplotlib	Visualization
Seaborn	Statistical plots
Jupyter Notebook	Implementation environment

4.7 System Architecture

The system follows a linear data-analytics workflow. Raw sales data is first ingested from a CSV file, passed through a cleaning and preprocessing stage to remove missing values, unwanted columns, and type inconsistencies, and then subjected to Exploratory Data Analysis. The EDA stage generates visual outputs (bar charts) across five dimensions — gender, age group, state, occupation, and product category — which feed into the final results and insight-generation stage.

- Data Source: Diwali Sales Dataset (CSV file)
- Processing Layer: Pandas / NumPy (cleaning, transformation)
- Analysis Layer: Grouping and aggregation (groupby, sum)

- Visualization Layer: Matplotlib / Seaborn
- Output: Charts and summarized insights

4.8 Methodology

The project follows a standard data analytics workflow, described below.

4.8.1 Data Collection

Dataset used: “Diwali Sales Dataset” (CSV file).

4.8.2 Data Cleaning

- Removing missing values
- Converting columns to appropriate data types
- Removing unnecessary columns (Status, unnamed columns, etc.)

4.8.3 Exploratory Data Analysis (EDA)

The following plots were created:

- Gender-wise purchase
- Age-group analysis
- State-wise total orders and sales
- Occupation contribution
- Product category analysis

4.8.4 Implementation Code

The core implementation, using Pandas, NumPy, Matplotlib, and Seaborn, is summarized below:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")

# Load dataset
df = pd.read_csv("Diwali Sales Data.csv", encoding='latin1')

# Data cleaning
df.dropna(inplace=True)
df.drop(['Status', 'unnamed1'], axis=1, inplace=True)
df['Age'] = df['Age'].astype(str)
df.reset_index(drop=True, inplace=True)

# Gender-wise sales
gender_sales = df.groupby('Gender')['Amount'].sum()
```



```
sns.barplot(x=gender_sales.index, y=gender_sales.values)

# Age group-wise sales
age_sales = df.groupby('Age Group')['Amount'].sum()
sns.barplot(x=age_sales.index, y=age_sales.values)

# State-wise sales (Top 10)
state_sales = df.groupby('State')['Amount'].sum().nlargest(10)
state_sales.plot(kind='bar')

# Occupation-wise sales
occupation_sales = df.groupby('Occupation')['Amount'].sum()
sns.barplot(x=occupation_sales.index, y=occupation_sales.values)

# Product category-wise sales
product_sales = df.groupby('Product Category')['Amount'].sum()
sns.barplot(x=product_sales.index, y=product_sales.values)
```

4.9 Expected Outcomes / Results

The analysis successfully identified clear patterns in customer purchasing behaviour during the Diwali festive season. Key results obtained were:

- Women, especially married women aged 26–35, were the highest contributors to Diwali sales, both in number of purchases and total revenue.
- The 26–35 age group recorded the highest overall spending, followed by the 36–45 age group.
- Uttar Pradesh, Maharashtra, and Karnataka emerged as the top revenue-generating states, with Mumbai, Bengaluru, and Lucknow as major shopping hubs.
- Customers working in IT, Healthcare, and Aviation sectors were among the highest spenders.
- Clothing & Apparel, Electronics & Gadgets, and Home Decor & Appliances were the most purchased product categories.

These findings demonstrate the value of data analytics in understanding festive-season consumer behaviour, and provide actionable direction for targeted marketing, inventory planning, and customer engagement strategies. Potential future enhancements include applying machine learning models for purchase prediction and customer segmentation, building interactive dashboards (Power BI/Tableau/Streamlit), market basket analysis, RFM analysis, and time-series forecasting of festive sales.

4.10 Certificates of Completion and Communication Mail Proof

The internship completion certificate issued by UpSkill Campus and UniConverge Technologies, along with the official communication mail confirming the certificate, are attached below as proof of completion.



Fig 4.1: Internship Completion Certificate

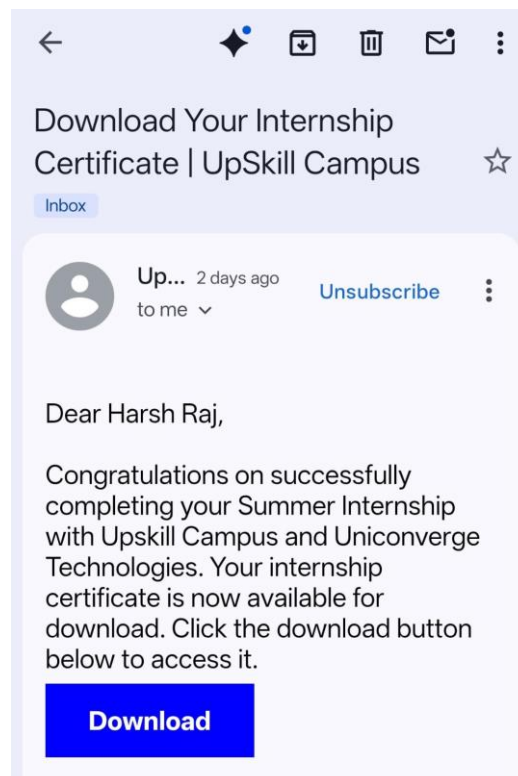


Fig 4.2: Certificate Communication Mail Proof

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